

Q1) 2)

Ass 08

Non-polar - Incompressible - Const μ

Variables $\rightarrow u, v, x, y, p, t$

$$u^* = \frac{u}{C} \quad v^* = \frac{v}{C} \quad x^* = \frac{x}{L} \quad y^* = \frac{y}{L} \quad t^* = \frac{tC}{L^*} \quad p^* = \frac{p}{\rho C^2}$$

Continuity Equation

$$\frac{\partial(u^* C)}{\partial(x^* L)} + \frac{\partial(v^* C)}{\partial(y^* L)} = 0 \quad \rightarrow * \frac{L}{C}$$

$$\rightarrow \frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0 \quad *$$

X-Momentum Equation

$$\rho \left(\frac{\partial(u^* C)}{\partial(t^* L)} + u^* C \frac{\partial(u^* C)}{\partial(x^* L)} + v^* C \frac{\partial(u^* C)}{\partial(y^* L)} \right) = - \frac{\partial(p^* \rho C^2)}{\partial(x^* L)} + \frac{\partial}{\partial x^*} \left(\mu C \frac{\partial(u^* C)}{\partial(x^* L)} \right) + \frac{\partial}{\partial y^*} \left(\mu C \left(\frac{\partial(u^* C)}{\partial(y^* L)} + \frac{\partial(v^* C)}{\partial(x^* L)} \right) \right)$$

$$\rightarrow \rho \left(\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} \right) = - \rho \frac{\partial p^*}{\partial x^*} + \frac{2\mu}{LC} \frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\mu}{CL} \frac{\partial}{\partial y^*} \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right)$$

$$\therefore \frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = - \frac{\partial p^*}{\partial x^*} + \frac{2}{Re} \frac{\partial^2 u^*}{\partial x^{*2}} + \frac{1}{Re} \frac{\partial}{\partial y^*} \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right)$$

$$\therefore \frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = - \frac{\partial p^*}{\partial x^*} + \frac{1}{Re} \left[2 \frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial}{\partial y^*} \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right] *$$

Y-Momentum

$$\rho \left[\frac{\partial (v^* c)}{\partial (t^* L)} + (u^* c) \frac{\partial (v^* c)}{\partial (x^* L)} + (v^* c) \frac{\partial (v^* c)}{\partial (y^* L)} \right] = - \frac{\partial (p^* c^2)}{\partial (y^* L)} + \frac{\partial}{\partial (y^* L)} \left(2\mu \frac{\partial v^* c}{\partial (y^* L)} \right) + \frac{\partial}{\partial (x^* L)} \left(\mu \left(\frac{\partial u^* c}{\partial (y^* L)} + \frac{\partial v^* c}{\partial (x^* L)} \right) \right)$$

$$\rho \left[\frac{\partial v^*}{\partial t^*} + u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} \right] = - \rho \frac{\partial p^*}{\partial y^*} + \frac{\partial}{\partial y^*} \left(2\mu \frac{\partial v^*}{\partial y^*} \right) + \frac{\partial}{\partial x^*} \left(\mu \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right)$$

$$\frac{\partial v^*}{\partial t^*} + u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = - \frac{\partial p^*}{\partial y^*} + \frac{2}{Re} \frac{\partial}{\partial y^*} \left(\frac{\partial v^*}{\partial y^*} \right) + \frac{1}{Re} \frac{\partial}{\partial x^*} \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right)$$

$$\frac{\partial v^*}{\partial t^*} + u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = - \frac{\partial p^*}{\partial y^*} + \frac{1}{Re} \left[2 \frac{\partial}{\partial y^*} \left(\frac{\partial v^*}{\partial y^*} \right) + \frac{\partial}{\partial x^*} \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right] \quad \#$$

ii) For Non-Newtonian Case

X-Momentum

$$\mu^* = \frac{\mu}{\rho c^2 \frac{L}{\rho}} = \frac{\mu}{\rho c L} \rightarrow \mu = \rho c L \mu^*$$

$$\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = - \frac{\partial p^*}{\partial x^*} + \frac{1}{\rho c} \frac{\partial}{\partial x^*} \left(2 \frac{\partial \mu^*}{\partial x^*} \mu^* \frac{\partial u^*}{\partial x^*} \right) + \frac{1}{\rho c} \frac{\partial}{\partial y^*} \left(\frac{\partial \mu^*}{\partial y^*} \mu^* \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right)$$

$$\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = - \frac{\partial p^*}{\partial x^*} + \frac{\partial}{\partial x^*} \left(2 \mu^* \frac{\partial u^*}{\partial x^*} \right) + \frac{\partial}{\partial y^*} \left(\mu^* \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right)$$

1- Momentum

$$\frac{\partial v^*}{\partial t^*} + u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = - \frac{\partial p^*}{\partial y^*} + \frac{\partial}{\partial y^*} \left(\frac{2}{Re} \mu^* \frac{\partial v^*}{\partial y^*} \right) + \frac{1}{Re} \frac{\partial}{\partial x^*} \left(\mu^* \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right)$$

$$\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = - \frac{\partial p^*}{\partial x^*} + \frac{\partial}{\partial x^*} \left(\frac{2}{Re} \mu^* \frac{\partial v^*}{\partial y^*} \right) + \frac{\partial}{\partial x^*} \left(\mu^* \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \right)$$

Shear dependent Viscosity

$$\mu = \mu_0 L \mu^* \rightarrow \dot{\gamma}^* = \frac{\dot{\gamma}}{L} = \frac{L}{C} \dot{\gamma} \rightarrow \dot{\gamma} = \frac{C}{L} \dot{\gamma}^*$$

$$\mu_0 L \mu^* = \mu_\infty + (\mu_0 - \mu_\infty) \left[1 + \left| \frac{bC}{L} \right|^a |\dot{\gamma}^*|^a \right]^{\frac{n-1}{a}}$$

$$\therefore \mu^* = \frac{1}{Re_\infty} + \left(\frac{1}{Re_0} - \frac{1}{Re_\infty} \right) \left[1 + \left| \frac{bC}{L} \right|^a |\dot{\gamma}^*|^a \right]^{\frac{n-1}{a}}$$

Remark $\rightarrow \dot{\gamma} b$ should be non dimensional $\rightarrow b$ has a unit of sec

Rate of Deformation tensor

$$\nabla \mathbf{V} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \quad \therefore [\nabla \mathbf{V}]^T = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{bmatrix}$$

$$\mathbf{D} = \frac{1}{2} (\nabla \mathbf{V} + [\nabla \mathbf{V}]^T) = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & \frac{\partial v}{\partial y} \end{bmatrix}$$

$$\therefore \mathbf{D}^* = \begin{bmatrix} \frac{\partial(u^*C)}{\partial(x^*L)} & \frac{1}{2} \left(\frac{\partial(u^*C)}{\partial(y^*L)} + \frac{\partial(v^*C)}{\partial(x^*L)} \right) \\ \frac{1}{2} \left(\frac{\partial(v^*C)}{\partial(x^*L)} + \frac{\partial(u^*C)}{\partial(y^*L)} \right) & \frac{\partial(v^*C)}{\partial(y^*L)} \end{bmatrix}$$

$$\therefore \mathbf{D}^* = \frac{C}{L} \begin{bmatrix} \frac{\partial u^*}{\partial x^*} & \frac{1}{2} \left(\frac{\partial u^*}{\partial y^*} + \frac{\partial v^*}{\partial x^*} \right) \\ \frac{1}{2} \left(\frac{\partial v^*}{\partial x^*} + \frac{\partial u^*}{\partial y^*} \right) & \frac{\partial v^*}{\partial y^*} \end{bmatrix} \quad \#$$

Remark $\rightarrow D_{ij} = \frac{C}{L} D_{ij}^*$

Shear Rate tensor

$$\dot{\gamma} = \sqrt{2 D_{ij} D_{ij}} \rightarrow \dot{\gamma}^2 = 2 \left[D_{11}^2 + D_{22}^2 + 2 D_{12} D_{21} \right]$$

$$\dot{\gamma} = \frac{c}{L} \dot{\gamma}^* \quad D_{ij} = \frac{c}{L} D_{ij}^*$$

$$\therefore \left(\frac{c}{L} \right)^2 (\dot{\gamma}^*)^2 = 2 \left[\left(\frac{c}{L} \right)^2 \left(\frac{\partial u^*}{\partial x^*} \right)^2 + \left(\frac{c}{L} \right)^2 \left(\frac{\partial v^*}{\partial y^*} \right)^2 + \left(\frac{c}{L} \right)^2 \left(\frac{\partial v^*}{\partial x^*} + \frac{\partial u^*}{\partial y^*} \right)^2 \right]$$

$$\therefore \dot{\gamma}^* = \sqrt{2 \left(\frac{\partial u^*}{\partial x^*} \right)^2 + 2 \left(\frac{\partial v^*}{\partial y^*} \right)^2 + \left(\frac{\partial v^*}{\partial x^*} + \frac{\partial u^*}{\partial y^*} \right)^2}$$

Question (a)

Unsteady - Compressible - 1D - Newtonian

Continuity equation

$$\rho^* = \frac{\rho}{\rho_\infty} \quad u^* = \frac{u}{c} \quad t^* = \frac{t c}{L} \quad x^* = \frac{x}{L} \quad c_\infty = \sqrt{\gamma R T_\infty}$$

$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} = 0$$

$$\rightarrow \frac{\partial (\rho^* c_\infty)}{\partial (t^* L)} + u^* c_\infty \frac{\partial (\rho^* c_\infty)}{\partial (x^* L)} + (\rho_\infty \rho^*) \frac{\partial (u^* c_\infty)}{\partial (x^* L)} = 0$$

$$\therefore \frac{\partial \rho^*}{\partial t^*} + u^* \frac{\partial \rho^*}{\partial x^*} + \rho^* \frac{\partial u^*}{\partial x^*} = 0 \quad //$$

X-Momentum Equation

$$\rho^* = \frac{\rho}{\rho_\infty c_\infty^2}$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} - \frac{\mu}{\rho} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = 0$$

$$\rightarrow \frac{\partial (u^* c_\infty)}{\partial (t^* L)} + (u^* c_\infty) \frac{\partial (u^* c_\infty)}{\partial (x^* L)} + \frac{1}{\rho^* \rho_\infty c_\infty^2} \frac{\partial (p^* \rho_\infty c_\infty^2)}{\partial (x^* L)} - \frac{\mu}{\rho^* \rho_\infty c_\infty^2} \frac{\partial}{\partial (x^* L)} \left(\frac{\partial (u^* c_\infty)}{\partial (x^* L)} \right) = 0$$

$$\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + \frac{1}{\rho^*} \frac{\partial p^*}{\partial x^*} - \frac{1}{\rho^*} \left[\frac{1}{\rho^*} \frac{\partial}{\partial x^*} \left(\frac{\partial u^*}{\partial x^*} \right) \right] \quad *$$

Energy Equation

$$\frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} + \gamma p \frac{\partial u}{\partial x} - \frac{4\mu(\gamma-1)}{3} \left(\frac{\partial u}{\partial x} \right)^2 = 0$$

$$\frac{\partial(p^* \rho_\infty^*)}{\partial(t^* L)} + (u^* \rho_\infty^*) \frac{\partial(p^* \rho_\infty^*)}{\partial(x^* L)} + \gamma(p^* \rho_\infty^*) \frac{\partial u^* \rho_\infty^*}{\partial(x^* L)} - \frac{4}{3}(\gamma-1) \left(\frac{\mu}{C_\infty} \right) \left(\frac{\partial u^*}{\partial(x^*)} \right)^2 \frac{\rho_\infty^*}{L} = 0$$

$$\frac{\partial p^*}{\partial t^*} + u^* \frac{\partial p^*}{\partial x^*} + \gamma p^* \frac{\partial u^*}{\partial x^*} - \frac{4(\gamma-1)}{3Re} \left(\frac{\partial u^*}{\partial x^*} \right)^2 = 0 \quad \text{**}$$

Energy Equation Derivation

For 1D case

$$\frac{\partial}{\partial t} \left[\rho \left(e + \frac{u^2}{2} \right) \right] + \frac{\partial}{\partial x} \left[\rho \left(e + \frac{u^2}{2} \right) u \right] = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) - \frac{\partial(\mu p)}{\partial x} + \frac{\partial(\mu \gamma_{xx})}{\partial x}$$

↑ Ignore

$$\gamma_{xx} = \frac{4}{3} \mu \frac{\partial u}{\partial x}$$

Assuming perfect gas $\rightarrow e = C_v T \quad \& \quad C_v = \frac{R}{\gamma-1}$

$$T = \frac{p}{\rho R}$$

$$\therefore e = \frac{R}{\gamma-1} \frac{p}{\rho R} = \frac{p}{(\gamma-1)\rho}$$

$$\frac{\partial}{\partial t} \left[\frac{\rho}{\gamma-1} + \rho \frac{u^2}{2} \right] + \frac{\partial}{\partial x} \left[\frac{\rho u^3}{\gamma-1} + \frac{\rho u^3}{2} + u p \right] = \frac{\partial (u \tau_w)}{\partial x}$$

$$\rightarrow \frac{\rho}{\gamma-1} + \left[\frac{\partial}{\partial t} \left(\rho \frac{u^2}{2} \right) + \frac{\partial}{\partial x} \left(\rho \frac{u^3}{2} \right) \right] + \frac{\partial}{\partial x} \left[u p \left(\frac{\gamma}{\gamma-1} \right) \right] = \frac{\partial (u \tau_w)}{\partial x}$$

from continuity

$$\rightarrow \rho u u_x + \frac{u^2}{2} \rho_x + \rho \frac{3}{2} u^2 u_x + \frac{u^3}{2} \rho_x \rightarrow \frac{u^2}{2} \left[\rho_x + 4 \rho_x + \rho u_x + 2 \rho u_x \right] + \rho u u_x$$

$$\rightarrow \rho u^2 u_x + \rho u u_x \rightarrow \rho u \left[u u_x + u_x \right] = \rho u \left[\frac{u}{2} \rho_x + \frac{u}{2} u_{xx} \right] \quad \text{from X-momentum}$$

$$\rightarrow \frac{4}{3} \rho u u_{xx} - u \rho_x$$

$$\rightarrow \rho_x + (\gamma-1) \left[\frac{4}{3} \rho u u_{xx} - u \rho_x \right] + \rho u \rho_x + \rho u_x - \frac{4 \rho (\gamma-1)}{3} \left[u u_{xx} + u_x^2 \right] = 0$$

$$\rightarrow \rho_x - \gamma u \rho_x + u \rho_x + \gamma u \rho_x + \rho u_x - \frac{4 \rho (\gamma-1)}{3} u_x^2 = 0$$

$$\rightarrow \rho_x + u \rho_x + \gamma \rho u_x - \frac{4 \rho (\gamma-1)}{3} u_x^2 = 0 \quad //$$

Question 12) (H)

$$\frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} + \gamma p \frac{\partial u}{\partial x} - \frac{4\gamma e}{3} (\gamma - 1) \left(\frac{\partial u}{\partial x} \right)^2 = 0$$

Card

$$\frac{p^{n+1} - p^n}{\Delta t} + u^n p_x^{n+1} + \gamma p^n u_x^{n+1} - \frac{4\gamma e (\gamma - 1)}{3} u_x^n u_x^{n+1} = 0$$

Newton-Raphson

$$p^{n+1} = p^n + \Delta p \quad u^{n+1} = u^n + \Delta u \quad \rightarrow \text{all variables at } n+1 \text{ time}$$

$$\rightarrow \frac{p^n + \Delta p - p^n}{\Delta t} + \underbrace{(u^n + \Delta u) \frac{\partial (p^n + \Delta p)}{\partial x}}_{\text{I}} + \underbrace{\gamma (p^n + \Delta p) \frac{\partial (u^n + \Delta u)}{\partial x}}_{\text{II}} - \underbrace{\frac{4\gamma e (\gamma - 1)}{3} \left(\frac{\partial u^n}{\partial x} + \frac{\partial \Delta u}{\partial x} \right)^2}_{\text{III}} = 0$$

$$\text{For I} \rightarrow u^n \frac{\partial p^n}{\partial x} + u^n \frac{\partial \Delta p}{\partial x} + \Delta u \frac{\partial p^n}{\partial x} + \cancel{\Delta u \frac{\partial \Delta p}{\partial x}}$$

$$\text{For II} \rightarrow \gamma p^n \frac{\partial u^n}{\partial x} + \gamma p^n \frac{\partial \Delta u}{\partial x} + \gamma \Delta p \frac{\partial u^n}{\partial x} + \cancel{\gamma \Delta p \frac{\partial \Delta u}{\partial x}}$$

$$\text{For III} \rightarrow A \left[\left(\frac{\partial u^n}{\partial x} \right)^2 + 2 \frac{\partial \Delta u}{\partial x} \frac{\partial u^n}{\partial x} + \left(\frac{\partial \Delta u}{\partial x} \right)^2 \right] = A \left(\frac{\partial u^n}{\partial x} \right)^2 + 2A \frac{\partial \Delta u}{\partial x} \frac{\partial u^n}{\partial x} \quad A = \frac{-4\gamma e (\gamma - 1)}{3}$$

$$\rightarrow \frac{p^n + \Delta p - p^n}{\Delta t} + p_x^n (u^n + \Delta u) + u_x^n (\Delta p)_x + \gamma \left[u_x^n (p^n + \Delta p) + p_x^n (\Delta u)_x \right] - \frac{4\gamma e (\gamma - 1)}{3} \left[(u_x^n)^2 + 2 u_x^n (\Delta u)_x \right] = 0$$

Question (3)

$$x^2 + 5x - 6 = 0$$

Picard

$$x^{k+1} x^k + 5x^k - 6 = 0 \rightarrow x^{k+1} = \frac{6 - 5x^k}{x^k} = \frac{6}{x^k} - 5$$

Newton Raphson

$$x^{k+1} = x^k + \Delta x$$

$$(x^k + \Delta x)^2 + 5(x^k + \Delta x) - 6 = 0$$

$$(x^k)^2 + 2x^k \Delta x + (\Delta x)^2 + 5x^k + 5\Delta x - 6 = 0$$

$$\therefore \Delta x = \frac{6 - 5x^k - (x^k)^2}{2x^k + 5}$$

$$u_t + u u_x = \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right)$$

Solution

$$R = u_t + u u_x - \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right) \quad \rightarrow W_i = N_i + \tau L(N_i)$$

$$L(1) = \frac{\partial(1)}{\partial t} + u \frac{\partial(1)}{\partial x} - \frac{\partial}{\partial x} \left(\mu \frac{\partial(1)}{\partial x} \right)$$

$$L(N_i) = \frac{\partial N_i}{\partial t} + u \frac{\partial N_i}{\partial x} - \frac{\partial}{\partial x} \left(\mu \frac{\partial N_i}{\partial x} \right) = u N_{ix} \quad \rightarrow \text{assuming const } \mu$$

$$W_i = N_i + \tau u N_{ix}$$

$$W_{ix} = N_{ix} + \tau (u N_{ixx} + N_{ix} u_x) = N_{ix} + \tau N_{ix} u_x$$

$$\rightarrow \int_{-1}^1 R W_i \cdot dx = 0 \rightarrow \int_{-1}^1 (u_t + u u_x - \mu u_{xx}) W_i \cdot dx = 0$$

$$\rightarrow \int_{-1}^1 (u_t W_i + u u_x W_i) - \mu \left[\underbrace{u_x W_i}_{\text{B.C.}} \right]_{-1}^1 + \int_{-1}^1 u_x W_{ix} = 0$$

$$\rightarrow \int_{-1}^1 u_t W_i + u_x [u W_i + \mu W_{ix}] \cdot dx = 0$$

$$u_t = \dot{u}_j N_j \quad \rightarrow \dot{u}_j = \frac{u^{n+1} - u^n}{\Delta t}$$

$$\rightarrow \int_{x_j}^{x_{j+1}} \underbrace{\dot{u}_j N_j W_i}_{m_{ij}} + u_x \underbrace{[u W_i + \mu W_{ix}]}_{r_{ij}} \cdot dx = 0$$

$$u^{n+1} m_{ij} = -\Delta t r_{ij} + u^n m_{ij}$$

Lec Example

$$\frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial x^2} = 0 \quad -1 \leq x \leq 1$$

B.C: $T(t, -1) = 0 \quad \rightarrow \quad T(t, 1) = 0$

I.C: $T(0, x) = 1 - x^2$

Solution

$$R = \frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial x^2} \quad W_i = N_i \quad \rightarrow \text{as there is no Convection}$$

$$\int_{-1}^1 R W_i \cdot dx = 0 \rightarrow \int_{-1}^1 \frac{\partial T}{\partial t} N_i \cdot dx - \int_{-1}^1 \frac{\partial^2 T}{\partial x^2} N_i \cdot dx = 0$$

$$\int_{-1}^1 \frac{\partial T}{\partial t} N_i \cdot dx - \left. \frac{\partial T}{\partial x} N_i \right|_{-1}^1 + \int_{-1}^1 \frac{\partial T}{\partial x} N_i' \cdot dx = 0$$

$$\int_{-1}^1 \frac{\partial T}{\partial t} N_i + \frac{\partial T}{\partial x} N_i' \cdot dx - \left. \frac{\partial T}{\partial x} N_i \right|_{-1}^1 = 0$$

$$T(t, x) = N_j(x) T_j(t) = T_j(t) N_j(x)$$

$$\int_{x_1}^{x_2} \dot{T}_j N_j N_i + T_j N_j' N_i' \cdot dx = 0$$

$$\left(\int_{x_1}^{x_2} N_i N_j \cdot dx \right) \dot{T}_j + \left(\int_{x_1}^{x_2} N_i' N_j' \cdot dx \right) T_j = 0$$

$$m_{ij} \dot{T}_j + k_{ij} T_j = 0$$

$$[K]_{2 \times 2} = \int_{x_1}^{x_2} [N]_{2 \times 1}^T [N]_{1 \times 2} dx$$

$$[M]_{2 \times 2} = \int_{x_1}^{x_2} [N]_{2 \times 1}^T [N]_{1 \times 2} dx$$

$$\therefore [M]_{2 \times 2} [\dot{T}]_{2 \times 1} + [K]_{2 \times 2} [T]_{2 \times 1} = 0$$

Using Euler Differencing for $[\dot{T}]_{2 \times 1}$

$$\therefore [\dot{T}]_{2 \times 1} = \frac{1}{\Delta t} \left\{ [T]_{2 \times 1}^{n+1} - [T]_{2 \times 1}^n \right\} \therefore [T]_{2 \times 1}^{n+1} - [T]_{2 \times 1}^n$$

$$\therefore \frac{1}{\Delta t} [M]_{2 \times 2} [T]_{2 \times 1}^{n+1} - \frac{1}{\Delta t} [M]_{2 \times 2} [T]_{2 \times 1}^n + [K]_{2 \times 2} [T]_{2 \times 1}^n = 0$$

$$\therefore [M]_{2 \times 2} [T]_{2 \times 1}^{n+1} + \left\{ -[M]_{2 \times 2} + \Delta t [K]_{2 \times 2} \right\} [T]_{2 \times 1}^n = 0$$

$$\therefore [M]_{2 \times 2} [T]_{2 \times 1}^{n+1} = [G]_{2 \times 2} [T]_{2 \times 1}^n = [RHS]_{2 \times 1} //$$

$$[G]_{2 \times 2} = [M]_{2 \times 2} - \Delta t [K]_{2 \times 2} = \int_{x_1}^{x_2} [N]_{2 \times 1}^T [N]_{1 \times 2} - \Delta t [N]_{2 \times 1}^T [N]_{1 \times 2} dx$$

$$RHS = [G]_{2 \times 2} [T]_{2 \times 1}^n \therefore [T]_{2 \times 1}^n = \frac{1 - [X^2]_{2 \times 1}}{[X^2]_{2 \times 1}}$$

Lecture Example

$$u_t + u u_x + \frac{1}{\rho} p_x - \frac{\mu}{\rho} u_{xx} = 0$$

Picard

$$\frac{u^{n+1} - u^n}{\Delta t} + u^n (u_x)^{n+1} + \frac{1}{\rho^{n+1}} p_x^{n+1} - \frac{\mu}{\rho^{n+1}} (u_{xx})^{n+1} = 0$$

Newton Raphson

$$u^{n+1} = u^n + \Delta u \quad \rho^{n+1} = \rho^n + \Delta \rho \quad \rho^{n+1} = \rho^n + \Delta \rho \quad \text{all variables at } n+1$$

$$\therefore \underbrace{\frac{(u^n + \Delta u) - u^n}{\Delta t}}_I + \underbrace{(u^n + \Delta u) \left(\frac{\partial (u^n + \Delta u)}{\partial x} \right)}_{II} + \underbrace{\frac{1}{(\rho^n + \Delta \rho)} \frac{\partial (p^n + \Delta p)}{\partial x}}_{III} - \underbrace{\frac{\mu}{(\rho^n + \Delta \rho)} \frac{\partial^2 (u^n + \Delta u)}{\partial x^2}}_{IV} = 0$$

$$\text{For I} \rightarrow u^n \frac{\partial u^n}{\partial x} + u^n \frac{\partial \Delta u}{\partial x} + \Delta u \frac{\partial u^n}{\partial x} + \Delta u \frac{\partial \Delta u}{\partial x}$$

$$\text{For II} \rightarrow (\rho^n + \Delta \rho)^{-1} = \frac{1}{\rho^n} \left(1 + \frac{\Delta \rho}{\rho^n} \right)^{-1} = \frac{1}{\rho^n} \left[1 - \frac{\Delta \rho}{\rho^n} + \left(\frac{\Delta \rho}{\rho^n} \right)^2 + \dots \right] = A(1 - A \Delta \rho) \quad A = \frac{1}{\rho^n}$$

$$\text{For III} \rightarrow (A - A^2 \Delta \rho) \left(\frac{\partial p^n}{\partial x} + \frac{\partial \Delta p}{\partial x} \right) = A \frac{\partial p^n}{\partial x} + A \frac{\partial \Delta p}{\partial x} - A^2 \Delta \rho \frac{\partial p^n}{\partial x} - A^2 \Delta \rho \frac{\partial \Delta p}{\partial x}$$

$$\text{For IV} \rightarrow (-\mu A + \mu A^2 \Delta \rho) \left(\frac{\partial^2 u^n}{\partial x^2} + \frac{\partial^2 \Delta u}{\partial x^2} \right) = -\mu A \frac{\partial^2 u^n}{\partial x^2} - \mu A \frac{\partial^2 \Delta u}{\partial x^2} + \mu A^2 \Delta \rho \frac{\partial^2 u^n}{\partial x^2} + \mu A^2 \Delta \rho \frac{\partial^2 \Delta u}{\partial x^2}$$

$$\rightarrow \frac{u^n + \Delta u - u^n}{\Delta t} + I + III + IV = 0$$

$$\rightarrow \frac{u^n + \Delta u - u^n}{\Delta t} + u_x^n (u^n + \Delta u) + u^n u_x + \frac{1}{\rho^n} \left[p_x^n \left(1 - \frac{\Delta \rho}{\rho^n} \right) + (\Delta p)_x \right] - \frac{\mu}{\rho^n} \left[u_{xx}^n \left(1 - \frac{\Delta \rho}{\rho^n} \right) + (\Delta u)_{xx} \right] = 0$$